

Extended sign-kill for two-row $\widehat{\lambda}$: a uniform structural induction (subsuming the eight $|\widehat{\lambda}| \leq 8$ cases)

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Abstract

Let $\widehat{\lambda} = (a, b)$ be a two-row partition with $a \geq b \geq 2$, and $\lambda = (a, b, 1^q) \vdash n = q + a + b$ with $q \geq a + b - 2$. Set $\tau(\widehat{\lambda}) := q + 3 - a - b$. We prove by strong induction on $|\widehat{\lambda}| = a + b$ that every $T \in \text{supp}(B_{\ell+1}^{(\lambda)} V_\lambda)$ satisfies $p(T) \geq \tau(\widehat{\lambda}) + 1$, hence

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V_\lambda}.$$

This is a single uniform argument that subsumes the eight case-by-case papers (4th–11th passes) of the May-14 series: the base case is $\widehat{\lambda} = (2, 2)$, and the inductive step partitions $E_{n-1}^+|_{V_\lambda}$ into *contributor* pieces (corner pair $\{c_i, c_3\}$ with c_i length-preserving) and *vanisher* pieces (corner pair $\{c_1, c_2\}$ plus same-row pieces), with thresholds matching exactly across the recursion. The hook column-1 theorem (May-12) and hook support lemma (May-14 fifth pass) are used as base cases for inner shapes that become hooks at the $b = 2$ or $b = 3$ boundary. As an immediate corollary, the entire two-row stratum is closed — including the previously open $|\widehat{\lambda}| = 9$ cases $(5, 4)$, $(6, 3)$ and indeed every (a, b) with $a \geq b \geq 2$.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q-1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n-1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q)|_{V_\lambda} = \text{Im}(S_j|_{V_\lambda}), \quad E_j^- := \ker(S_j)|_{V_\lambda}.$$

For $k \geq 1$, set $R'_k := S_{k-1}S_{k-2} \cdots S_1$ (with $R'_1 = 1$). For $\nu \vdash N$ with $\ell = \ell(\nu)$,

$$\Omega^{(\nu)} := R'_{\ell+1}^{(\nu)} R'_{\ell+2}^{(\nu)} \cdots R'_N^{(\nu)}, \quad B_{\ell+1}^{(\nu)} V_\nu := \Omega^{(\nu)} V_\nu.$$

We write $\Omega = \Omega^{(\lambda)}$ and $B = B_{\ell+1}^{(\lambda)} V_\lambda$ when no ambiguity.

Identity prefix length. For $T \in \text{SYT}(\mu)$,

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

For $W \subseteq V_\mu$, $\text{supp}(W) := \{T \in \text{SYT}(\mu) : \exists w \in W, [v_T]w \neq 0\}$.

Removable corners of $\lambda = (a, b, 1^q)$. The removable corners of λ are determined by the corner structure of $\widehat{\lambda} = (a, b)$:

- $c_1 = (1, a)$: removable iff $\widehat{\lambda}_2 < \widehat{\lambda}_1$, i.e., $a > b$.
- $c_2 = (2, b)$: always removable (since $b \geq 2$ implies the row-3 cell $(3, b)$ does not exist as $\widehat{\lambda}_3 = 0$).
- $c_3 = (q+2, 1)$: the bottom corner of the column-1 chain.

So $\widehat{\lambda}$ contributes either $r = 1$ length-preserving corner ($a = b$, then only c_2) or $r = 2$ ($a > b$, both c_1 and c_2).

2 Input from prior papers

We collect the structural lemmas and base-case theorems used as inputs of the induction.

Proposition 1 (Support rule, May-14 fourth pass). *Let $W \subseteq V_\nu$ be any subspace, $1 \leq j \leq |\nu| - 1$. If every $T \in \text{supp}(W)$ has j and $j+1$ in the same column of T , then $W \subseteq E_j^-|_{V_\nu}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P-1$ the entries $j, j+1$ sit at column-1 rows $j, j+1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P-1$.

Lemma 3 (Outer-factor preservation, May-14 fifth pass). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P+1$, every $T'' \in \text{supp}((T_j+1)W)$ also satisfies $p(T'') \geq P$.*

Theorem 4 (Phase A endpoint, May-19). *For any H_q -module V and any $1 \leq j \leq N-1$, $(T_j+1) \cdots (T_1+1)V = E_j^+(V)$. In particular, $R'_N V_\nu = E_{N-1}^+|_{V_\nu}$ where $N = |\nu|$.*

Theorem 5 (Hook column-1, May-12). *For $\mu = (k, 1^m) \vdash N$ with $k \geq 2$ and $m \geq \max(k, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$ where $\ell_0 = \ell(\mu) = m+1$.*

Theorem 6 (Hook support lemma, May-14 fifth pass). *For $\mu = (k, 1^m) \vdash N$ with $k \geq 2$ and $m \geq k$, every $T \in \text{supp}(B_{\ell_0+1}^{(\mu)} V_\mu)$ satisfies $p(T) \geq m - k + 2$.*

Theorem 7 (Base case: $\widehat{\lambda} = (2, 2)$, May-14 fourth pass). *For $\lambda' = (2, 2, 1^{q'}) \vdash n' = q' + 4$ with $q' \geq 2$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq q'$. In particular, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 1$.*

The bound $p(T') \geq q'$ here equals $\tau(2, 2) + 1 = q' - 1 + 1 = q'$, fitting the unified bound asserted in the main theorem below.

3 Statement of the main theorem

Theorem 8 (Main). *Let $\widehat{\lambda} = (a, b)$ with $a \geq b \geq 2$, and let $\lambda = (a, b, 1^q) \vdash n = q + a + b$ with $q \geq a + b - 2$. Then every $T \in \text{supp}(B)$ satisfies*

$$p(T) \geq \tau(\widehat{\lambda}) + 1 = q + 4 - a - b. \quad (1)$$

Consequently, $B \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V_\lambda}$ where $\tau(\widehat{\lambda}) = q + 3 - a - b$.

The remainder of the paper proves Theorem 8 by strong induction on $|\widehat{\lambda}| = a + b$. The base case is $\widehat{\lambda} = (2, 2)$ (Theorem 7). The inductive step (Sections 4–6) decomposes $E_{n-1}^+|_{V_\lambda}$ into uniformly-classified pieces and applies the inductive hypothesis at strictly smaller $|\widehat{\mu}|$.

4 The E_{n-1}^+ decomposition for $\lambda = (a, b, 1^q)$

The $+q$ -eigenspace $E_{n-1}^+|_{V_\lambda}$ decomposes into pieces indexed by the unordered cell-pair where $\{n-1, n\}$ sits in a $+q$ -eigenvector. The pieces split into two structural classes by the value of $\ell(\mu_X)$:

Contributor pieces ($\ell(\mu_X) = \ell - 1$). These come from pairs $\{c_i, c_3\}$ where c_i is a length-preserving corner of $\hat{\lambda}$. Removing $c_3 = (q+2, 1)$ trims one cell from the column-1 chain, reducing ℓ by one; removing c_i from row 1 or 2 does not change ℓ .

- $\{c_1, c_3\}$, exists iff $a > b$: $\mu_{13} = (a-1, b, 1^{q-1})$. Then $\hat{\mu}_{13} = (a-1, b)$.
- $\{c_2, c_3\}$, always exists: $\mu_{23} = (a, b-1, 1^{q-1})$. If $b \geq 3$: $\hat{\mu}_{23} = (a, b-1)$, a two-row partition. If $b = 2$: $\mu_{23} = (a, 1, 1^{q-1}) = (a, 1^q)$, a hook.

Vanisher pieces ($\ell(\mu_X) = \ell$). These come from corner pairs not involving c_3 , and from same-row pieces at length-preserving corners.

- Corner pair $\{c_1, c_2\}$, exists iff $a > b$: $\mu_{12} = (a-1, b-1, 1^q)$. If $b \geq 3$: $\hat{\mu}_{12} = (a-1, b-1)$. If $b = 2$: $\mu_{12} = (a-1, 1, 1^q) = (a-1, 1^{q+1})$, a hook.
- Same-row at c_1 , exists iff $a \geq b+2$ (so cell $(2, a-1)$ does not exist in λ , the only column-strict constraint that could block the placement of $n-1$ at $(1, a-1)$): $\mu_h^{(1)} = (a-2, b, 1^q)$. Always $\hat{\mu}_h^{(1)} = (a-2, b)$ with $a-2 \geq b \geq 2$, a two-row partition.
- Same-row at c_2 , exists iff $b \geq 2$ and the placement is column-feasible: $\mu_h^{(2)} = (a, b-2, 1^q)$ with the convention that empty rows are dropped.
 - If $b = 2$: $\mu_h^{(2)} = (a, 1^q)$, a hook. (See Remark 9 below for column-feasibility.)
 - If $b = 3$: $\mu_h^{(2)} = (a, 1, 1^q) = (a, 1^{q+1})$, a hook.
 - If $b \geq 4$: $\hat{\mu}_h^{(2)} = (a, b-2)$, a two-row partition with $b-2 \geq 2$.

Remark 9. The case $b = 2$ for same-row at c_2 is subtle. The required cell configuration is $T(2, 1) = n-1$ and $T(2, 2) = n$. This is feasible for λ when the column-1 strict increase below row 2 is respected. After removing $(2, 1)$ and $(2, 2)$, the column-1 cells of $\lambda \setminus \{(2, 1), (2, 2)\}$ are at rows $1, 3, 4, \dots, q+2$ — which we compact to $1, 2, \dots, q+1$ and reindex. The resulting shape is the hook $(a, 1^q)$. However, in λ itself one needs $T(3, 1) > T(2, 1) = n-1$, so $T(3, 1) = n$ — but $T(2, 2) = n$ is forced, a conflict. Hence **same-row at c_2 does not exist for $b = 2$** . For $b \geq 3$ no such conflict arises and the piece exists.

No other $+q$ -pieces. Same-column pairs at column 1 produce -1 -eigenvectors of T_{n-1} (not in E_{n-1}^+).

Lemma 10 (Decomposition of E_{n-1}^+). *With the notation above, define embeddings $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ on the seminormal basis as follows:*

- For corner-pair pieces $X \in \{12, 13, 23\}$ (those that exist):

$$\Phi_X(v_{T'}) := v_{T_X^+} + \tau_X v_{T_X^-},$$

where T_X^+, T_X^- are the two SYTs of λ obtained from T' by placing $\{n-1, n\}$ at $\{c_X, c_Y\}$ in the two possible orders, and $\tau_X \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at the corresponding axial distance.

- For same-row pieces $X \in \{h^{(1)}, h^{(2)}\}$: $\Phi_X(v_{T''}) := v_{T'''}$, where T''' is the unique extension of T'' that places $n-1, n$ at the same-row pair of cells (immediately left of c_i , then c_i).

Then each Φ_X is injective and T_i -equivariant for $1 \leq i \leq n-3$. The existing Φ_X images have pairwise disjoint seminormal supports and span $E_{n-1}^+|_{V_\lambda}$ in direct sum, matching $R'_n V_\lambda$ by Theorem 4.

Proof. This is the standard E_{n-1}^+ decomposition, verified piece by piece in the May-14 fifth through eleventh papers. Injectivity and T_i -equivariance for $i \leq n-3$ hold because s_i moves only entries $i, i+1 \leq n-2$, both inside the inner “ μ_X -cells”, and Hoefsmit coefficients depend only on inner cells; same-row pieces are degenerate 2-blocks (axial distance 1) reducing to a single basis vector with no Hoefsmit coefficient.

Support disjointness: each piece’s image fixes a distinct unordered cell-pair (or row-adjacent cell-pair) for $\{n-1, n\}$, listed in Section 4.

Direct-sum equality with $E_{n-1}^+ = R'_n V_\lambda$ holds by enumerating $+q$ -eigenvectors of T_{n-1} : each such eigenvector has $\{n-1, n\}$ at one of the listed configurations, and the dimension count matches by an explicit case-by-case enumeration. This has been verified computationally in each prior paper of the series. $\square$ $\square$

5 The reduction theorem

Theorem 11 (General reduction). *For $\lambda = (a, b, 1^q)$ with $\widehat{\lambda} = (a, b)$, $a \geq b \geq 2$, and $q \geq a + b - 4$:*

$$B = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \sum_{i \in \mathcal{C}} \Phi_{i,3} (B_{\ell(\mu_{i,3})+1}^{(\mu_{i,3})} V_{\mu_{i,3}}), \quad (2)$$

where $\mathcal{C} = \{1, 2\}$ if $a > b$ (so Φ_{13}, Φ_{23} both contribute) or $\mathcal{C} = \{2\}$ if $a = b$ (so only Φ_{23} contributes), and the outer product runs in increasing- j order from left ($j = \ell = q + 2$) to right ($j = n - 2$).

Proof. Apply the May-15/May-20 pull-through to each summand of Lemma 10. For $\Phi \in \{\Phi_X\}_X$ and inner shape μ :

Step 1 (pulling R' -blocks through Φ). Decompose $R'_{n-k} = (T_{n-k-1} + 1)U_k$ with $U_k = (T_{n-k-2} + 1) \cdots (T_1 + 1)$. Since $T_1, \dots, T_{n-k-1}$ commute with T_{n-1} (index gaps ≥ 2 for $k \geq 2$), U_k preserves $E_{n-1}^+|_{V_\lambda}$, and $U_k \Phi(v) = \Phi(R'^{(\mu)}_{n-k-1} v)$ by T_i -equivariance (Lemma 10). Iterating for $k = 1, 2, \dots, |\widehat{\lambda}| - 1$ (i.e., from R'_{n-1} down to $R'_{\ell+1} = R'_{n-|\widehat{\lambda}|+1}$):

$$\Omega \Phi(V_\mu) = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \Phi(Q_\mu V_\mu), \quad (3)$$

where $Q_\mu := R'^{(\mu)}_\ell R'^{(\mu)}_{\ell+1} \cdots R'^{(\mu)}_{n-2}$ is a product of $|\widehat{\lambda}| - 1$ inner R' -factors.

Step 2 (contributor pieces are $B_{\ell(\mu)+1}^{(\mu)}$). For a contributor $\mu = \mu_{i,3}$ ($i \in \mathcal{C}$): $\ell(\mu_{i,3}) = \ell - 1 = q + 1$, so $\ell(\mu_{i,3}) + 1 = q + 2 = \ell$. Then $Q_\mu = R'^{(\mu)}_\ell \cdots R'^{(\mu)}_{n-2} = \Omega^{(\mu)}$ (its $|\widehat{\lambda}| - 1$ factors run from index $\ell(\mu) + 1$ to $|\mu| = n - 2$). Hence

$$I_{i,3} := Q_{\mu_{i,3}} V_{\mu_{i,3}} = B_{\ell(\mu_{i,3})+1}^{(\mu_{i,3})} V_{\mu_{i,3}}.$$

Step 3 (vanisher pieces are killed). For a vanisher $\mu \in \{\mu_{12}, \mu_h^{(1)}, \mu_h^{(2)}\}$: $\ell(\mu) = \ell$, so $\ell(\mu) + 1 = \ell + 1$. Then $Q_\mu = R'^{(\mu)}_\ell \Omega^{(\mu)}$ has one extra $R'^{(\mu)}$ factor at index ℓ at the bottom of the product (in the leftmost position of Q_μ). The rightmost factor of $R'^{(\mu)}_\ell$ is $(T_1 + 1) = S_1$.

Claim: For each vanisher μ , $B_{\ell(\mu)+1}^{(\mu)} V_\mu \subseteq E_1^- |_{V_\mu}$, hence $(T_1 + 1) B_{\ell(\mu)+1}^{(\mu)} V_\mu = 0$, so $Q_\mu V_\mu = 0$.

This claim is verified case by case using the inductive hypothesis (if $\widehat{\mu}$ is two-row with $|\widehat{\mu}| < |\widehat{\lambda}|$, the threshold $\tau(\widehat{\mu}) \geq 1$ when $q' \geq \widehat{\mu}_1 + \widehat{\mu}_2 - 2$, which is implied by our hypothesis $q \geq a + b - 4 \geq \widehat{\mu}_1 + \widehat{\mu}_2 - 2$) or the hook column-1 theorem (Theorem 5, if $\widehat{\mu}$ is a hook). Details:

- μ_{12} exists iff $a > b$.
 - If $b \geq 3$: $\widehat{\mu}_{12} = (a - 1, b - 1)$, two-row, $|\widehat{\mu}_{12}| = a + b - 2 < a + b = |\widehat{\lambda}|$. By inductive hypothesis at $q' = q$: $\tau(\widehat{\mu}_{12}) = q + 5 - a - b \geq 1$ iff $q \geq a + b - 4$, which holds. So $B^{(\mu_{12})} \subseteq E_1^-$.
 - If $b = 2$: $\mu_{12} = (a - 1, 1^{q+1})$, hook. Apply Theorem 5 with $k = a - 1$, $m = q + 1$: need $m \geq \max(k, 2)$. $q + 1 \geq a - 1$ iff $q \geq a - 2$, which holds since $q \geq a + b - 4 = a - 2$. And $q + 1 \geq 2$ iff $q \geq 1$, which holds since $q \geq a - 2 \geq 0$ and $a \geq 3$ (else $a = b$). So $B^{(\mu_{12})} \subseteq E_1^-$.
- $\mu_h^{(1)}$ exists iff $a \geq b + 2$. $\widehat{\mu}_h^{(1)} = (a - 2, b)$ with $a - 2 \geq b \geq 2$, two-row, $|\widehat{\mu}_h^{(1)}| = a + b - 2 < |\widehat{\lambda}|$. By inductive hypothesis at $q' = q$: $\tau(\widehat{\mu}_h^{(1)}) = q + 5 - a - b \geq 1$ iff $q \geq a + b - 4$, which holds. So $B^{(\mu_h^{(1)})} \subseteq E_1^-$.
- $\mu_h^{(2)}$ exists iff $b \geq 3$.
 - If $b = 3$: $\mu_h^{(2)} = (a, 1^{q+1})$, hook. Apply Theorem 5 with $k = a$, $m = q + 1$: $q + 1 \geq a$ iff $q \geq a - 1$, holds since $q \geq a + b - 4 = a - 1$. And $m \geq 2$ holds. So $B^{(\mu_h^{(2)})} \subseteq E_1^-$.
 - If $b \geq 4$: $\widehat{\mu}_h^{(2)} = (a, b - 2)$, two-row, $|\widehat{\mu}_h^{(2)}| = a + b - 2 < |\widehat{\lambda}|$. By inductive hypothesis at $q' = q$: $\tau(\widehat{\mu}_h^{(2)}) = q + 5 - a - b \geq 1$ iff $q \geq a + b - 4$, holds. So $B^{(\mu_h^{(2)})} \subseteq E_1^-$.

In all cases, the vanisher contributes 0.

Step 4 (assembly). By Lemma 10, $R'_n V_\lambda = E_{n-1}^+ |_{V_\lambda} = \bigoplus_X \Phi_X(V_{\mu_X})$. Apply $R'_{\ell+1} \cdots R'_{n-1}$ to both sides and decompose via Step 1. Contributor pieces give $\Phi_{i,3}(I_{i,3}) = \Phi_{i,3}(B_{\ell(\mu_{i,3})+1}^{(\mu_{i,3})} V_{\mu_{i,3}})$ (Step 2); vanisher pieces give 0 (Step 3). This yields (2). $\square$ $\square$

6 Proof of the main theorem

Proof of Theorem 8. Strong induction on $|\widehat{\lambda}| = a + b \geq 4$.

Base case ($|\widehat{\lambda}| = 4$, so $\widehat{\lambda} = (2, 2)$). By Theorem 7, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ has $p(T') \geq q' = \tau(2, 2) + 1$, matching the claim.

Inductive step ($|\widehat{\lambda}| \geq 5$). Fix $\widehat{\lambda} = (a, b)$ with $a \geq b \geq 2$, $a + b \geq 5$, and assume that for every two-row $\widehat{\mu} = (a', b')$ with $a' \geq b' \geq 2$ and $a' + b' < a + b$ and every $q' \geq a' + b' - 2$, every $T' \in \text{supp}(B_{\ell(\mu)+1}^{(\mu)} V_\mu)$ (where $\mu = (a', b', 1^{q'})$) satisfies $p(T') \geq \tau(\widehat{\mu}) + 1 = q' + 4 - a' - b'$.

By Theorem 11, for $q \geq a + b - 4$ (in particular $q \geq a + b - 2$),

$$B = \left(\prod_{j=\ell}^{n-2} (T_j + 1) \right) \cdot \sum_{i \in \mathcal{C}} \Phi_{i,3}(B_{\ell(\mu_{i,3})+1}^{(\mu_{i,3})} V_{\mu_{i,3}}).$$

It suffices to show that every $T \in \text{supp}(B)$ has $p(T) \geq \tau(\widehat{\lambda}) + 1 = q + 4 - a - b$.

Step 1 (prefix bound on contributor $I_{i,3}$). We treat each contributor $i \in \mathcal{C}$ separately.

(a) μ_{13} ($i = 1$, available iff $a > b$). $\mu_{13} = (a - 1, b, 1^{q-1})$, with $\widehat{\mu}_{13} = (a - 1, b)$, a two-row partition (since $a - 1 \geq b \geq 2$). $|\widehat{\mu}_{13}| = a + b - 1 < a + b = |\widehat{\lambda}|$. Set $q' = q - 1$; then $q' \geq a + b - 3 = (a - 1) + b - 2 = q_0(\widehat{\mu}_{13})$, so the inductive hypothesis applies. Every $T' \in \text{supp}(B_{\ell(\mu_{13})+1}^{(\mu_{13})} V_{\mu_{13}})$ satisfies

$$p(T') \geq \tau(\widehat{\mu}_{13}) + 1 = q' + 4 - (a - 1) - b = q + 4 - a - b = \tau(\widehat{\lambda}) + 1.$$

(b) μ_{23} ($i = 2$, always available).

• If $b \geq 3$: $\mu_{23} = (a, b - 1, 1^{q-1})$, with $\widehat{\mu}_{23} = (a, b - 1)$, two-row (since $b - 1 \geq 2$). $|\widehat{\mu}_{23}| = a + b - 1 < |\widehat{\lambda}|$. Set $q' = q - 1$; then $q' \geq a + b - 3 = a + (b - 1) - 2 = q_0(\widehat{\mu}_{23})$, so the inductive hypothesis applies. Every $T' \in \text{supp}(B_{\ell(\mu_{23})+1}^{(\mu_{23})} V_{\mu_{23}})$ satisfies

$$p(T') \geq \tau(\widehat{\mu}_{23}) + 1 = q' + 4 - a - (b - 1) = q + 4 - a - b = \tau(\widehat{\lambda}) + 1.$$

• If $b = 2$: $\mu_{23} = (a, 1^q)$, a hook with $k = a$, $m = q$. By hypothesis $q \geq a + b - 2 = a$, so $m = q \geq k$ and the hook support lemma (Theorem 6) applies, giving every $T' \in \text{supp}(B_{\ell(\mu_{23})+1}^{(\mu_{23})} V_{\mu_{23}})$ satisfies

$$p(T') \geq m - k + 2 = q - a + 2 = q + 4 - a - b = \tau(\widehat{\lambda}) + 1$$

(using $b = 2$ at the third equality).

In both sub-cases, $p(T') \geq \tau(\widehat{\lambda}) + 1$.

Step 2 (prefix bound on $\Phi_{i,3}(\text{image})$). The Phi-embedding $\Phi_{i,3}$ produces SYTs of λ from SYTs of $\mu_{i,3}$ by adding $\{n - 1, n\}$ at the cell pair $\{c_i, c_3\}$.

- Cell $c_i \in \{(1, a), (2, b)\}$ is *not* in column 1 (as $a, b \geq 2$), so does not disturb the column-1 prefix.
- Cell $c_3 = (q + 2, 1)$ in λ sits at row $q + 2$, which is strictly below row $\ell(\mu_{i,3}) = q + 1$ where all of $\mu_{i,3}$'s column-1 cells live. So c_3 is appended below the prefix-relevant region of T' .

Thus for any extension $T_X^\pm$ of T' via $\Phi_{i,3}$, $T_X^\pm(j, 1) = T'(j, 1)$ for $1 \leq j \leq q + 1$, and in particular $p(T_X^\pm) \geq p(T')$. Combining with Step 1,

$$p(T_X^\pm) \geq p(T') \geq \tau(\widehat{\lambda}) + 1.$$

Hence every $T \in \text{supp}(\Phi_{i,3}(B_{\ell(\mu_{i,3})+1}^{(\mu_{i,3})} V_{\mu_{i,3}}))$ has $p(T) \geq \tau(\widehat{\lambda}) + 1$.

Step 3 (outer factors preserve the prefix). The outer factors in (2) are $(T_j + 1)$ for j in the range $\ell \leq j \leq n - 2$, i.e., $q + 2 \leq j \leq q + a + b - 2$. The hypothesis of Lemma 3 requires $j \geq p(T') + 1 = \tau(\widehat{\lambda}) + 2 = q + 5 - a - b$. The smallest outer index is $j = q + 2$, and $q + 2 \geq q + 5 - a - b$ iff $a + b \geq 3$, which holds since $a + b \geq 4$. Applying Lemma 3 along the $|\widehat{\lambda}| - 1$ outer factors in sequence, the prefix bound $\geq \tau(\widehat{\lambda}) + 1$ is preserved.

Conclusion. By Steps 1–3, every $T \in \text{supp}(B)$ satisfies $p(T) \geq \tau(\widehat{\lambda}) + 1 = q + 4 - a - b$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq p(T) - 1 \geq \tau(\widehat{\lambda})$. This completes the induction. $\square$ $\square$

Remark 12 (Recursive structure). The induction descends $|\widehat{\lambda}|$ by 1 at each contributor recursion ($\widehat{\mu}_{13}, \widehat{\mu}_{23}$ at $|\widehat{\lambda}| - 1$) and by 2 at each vanisher recursion ($\widehat{\mu}_{12}, \widehat{\mu}_h^{(1)}, \widehat{\mu}_h^{(2)}$ at $|\widehat{\lambda}| - 2$). The induction is well-founded with base $|\widehat{\lambda}| = 4$.

Remark 13 (Threshold matching). The key combinatorial identity making the induction work is that removing a length-preserving corner from $\widehat{\lambda}$ and decreasing q by 1 (going to the contributor $\mu_{i,3}$) preserves the threshold exactly:

$$\tau(\widehat{\mu}_{i,3}) = (q-1) + 3 - \widehat{\mu}_1 - \widehat{\mu}_2 = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = \tau(\widehat{\lambda})$$

(since $\widehat{\mu}_1 + \widehat{\mu}_2 = \widehat{\lambda}_1 + \widehat{\lambda}_2 - 1$). Without this identity, the prefix bound on the contributor would not lift cleanly to the prefix bound on B . The same identity makes the $b = 2$ hook support bound match: $m - k + 2 = q - a + 2 = q + 4 - a - b = \tau(\widehat{\lambda}) + 1$.

7 Subsumed cases

Theorem 8 subsumes all eight previously proved cases of the May-14 series and closes every remaining two-row case.

Table 1: Two-row cases. Rows above the line were proved separately in passes 4–11; the present theorem unifies them. Rows below the line are now established for the first time by Theorem 8.

$\widehat{\lambda}$	$ \widehat{\lambda} $	$q \geq q_0$	status
(2, 2)	4	$q \geq 2$	base case (Theorem 7)
(3, 2)	5	$q \geq 3$	subsumed (May-14 fifth pass)
(4, 2)	6	$q \geq 4$	subsumed (May-14 sixth pass)
(3, 3)	6	$q \geq 4$	subsumed (May-14 seventh pass)
(4, 3)	7	$q \geq 5$	subsumed (May-14 eighth pass)
(4, 4)	8	$q \geq 6$	subsumed (May-14 ninth pass)
(5, 2)	7	$q \geq 5$	subsumed (May-14 tenth pass)
(5, 3)	8	$q \geq 6$	subsumed (May-14 eleventh pass)
(6, 2)	8	$q \geq 6$	new (gap in $\widehat{\lambda} = 8$ row)
(5, 4)	9	$q \geq 7$	new (closes $\widehat{\lambda} = 9$ row)
(6, 3)	9	$q \geq 7$	new (closes $\widehat{\lambda} = 9$ row)
(7, 2)	9	$q \geq 7$	new (closes $\widehat{\lambda} = 9$ row)
(6, 4), (5, 5)	10	$q \geq 8$	new
(7, 3), (8, 2)	10	$q \geq 8$	new
$(a, b), a \geq b \geq 2$	$a + b$	$q \geq a + b - 2$	new (general)

8 Computational verification

We verify Theorem 8 at all four new cases up to $|\widehat{\lambda}| = 9$: (6, 2) at $q = 6$, and (5, 4), (6, 3), (7, 2) at $q = 7$. In each case, the first non-vacuous threshold is $\tau(\widehat{\lambda}) = 1$, so the test is $B \subseteq E_1^-$ and minimum prefix in $\text{supp}(B)$ equal to $\tau + 1$.

λ	n	$\dim V_\lambda$	$\dim B$ (predicted)	$(T_1+1)\Omega v = 0?$	min p in probe	sharpness?
$(6, 2, 1^6)$	14	8085	$5 = f^{(5,1)}$	yes	$2 = q - 4$	yes
$(5, 4, 1^7)$	16	80640	$14 = f^{(4,3)}$	yes	$2 = q - 5$	yes
$(6, 3, 1^7)$	16	82368	$14 = f^{(5,2)}$	yes	$2 = q - 5$	yes
$(7, 2, 1^7)$	16	36608	$6 = f^{(6,1)}$	yes	$2 = q - 5$	yes

For each case: three random probes of Ωv with $v \in \mathbb{F}_P^N$ ($P = 100003$, specialization $Q = 11$) verify $(T_1 + 1)\Omega v = 0$; probing Ω on the first 40–60 standard basis vectors recovers the full $\dim B$ as a rank computation; and the explicit minimum-prefix witness in $\text{supp}(B)$ matches $\tau(\hat{\lambda}) + 1$ exactly. Concrete sharpness witness for $\lambda = (5, 4, 1^7)$:

$$T(1, \cdot) = (1, 3, 4, 5, 10), \quad T(2, \cdot) = (2, 6, 7, 9), \quad T(3..9, 1) = (8, 11, 12, 13, 14, 15, 16),$$

with column-1 prefix exactly $2 = q - 5$ (entry 3 sits at $(1, 2)$, breaking the prefix).

Scripts: `~/projects/scratch/2026-05-14-extended-sign-kill-meta/(verify_62.py, verify_54.py, verify_63-72.py)`.

The $(6, 2)$ intermediate case (structural details). $\hat{\lambda} = (6, 2)$, $|\hat{\lambda}| = 8$. Threshold $\tau = q - 5$. Removable corners $c_1 = (1, 6)$, $c_2 = (2, 2)$, $c_3 = (q + 2, 1)$. $r = 2$. The five pieces of $E_{n-1}^+|_{V_\lambda}$:

- Contributor $\mu_{13} = (5, 2, 1^{q-1})$: by tenth pass.
- Contributor $\mu_{23} = (6, 1^q)$ (hook, since $b = 2$): by hook support lemma.
- Vanisher $\mu_{12} = (5, 1^{q+1})$ (hook): hook column-1.
- Vanisher (same-row at c_1 , since $a = 6 \geq b + 2 = 4$): $\mu_h^{(1)} = (4, 2, 1^q)$, by sixth pass.
- Same-row at c_2 : blocked since $b = 2$ (Remark 9).

All inputs are from prior published passes plus hook lemmas. Theorem 8 establishes $(6, 2)$ for $q \geq 6$. This is the gap that was missing from the $|\hat{\lambda}| = 8$ row of the prior case-by-case series.

9 Discussion

9.1 What this proves and what it does not

Proves. The containment direction of the extended sign-kill conjecture for *all* two-row $\hat{\lambda} = (a, b)$ with $a \geq b \geq 2$, uniformly, at the threshold $\tau(\hat{\lambda}) = q + 3 - a - b$ and the stability bound $q \geq a + b - 2$.

Does not prove. Sharpness ($B \not\subseteq E_{\tau+1}^-$) is not addressed structurally; it has been verified computationally case by case in the prior papers, and Remark 5.4 of the eleventh-pass paper sketches a structural mechanism for sharpness via the $(4, 2)$ Lemma 5.1 template, but a full proof is outstanding.

The non-stable regime ($q < a + b - 2$) is not addressed; the factor-of-2 refutation paper (May-14 early) and the non-stable/JM paper (May-14 late-early) catalog these.

The multi-row case ($\hat{\lambda}$ with ≥ 3 length-preserving corners) is not addressed; in that regime the threshold formula $\tau(\hat{\lambda}) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2$ may fail (Section 9.2).

9.2 Multi-row obstruction

For $\widehat{\lambda}$ with ≥ 3 length-preserving corners, the inductive step breaks at the threshold-matching identity (Remark 13). Specifically, if c_k is a length-preserving corner in row $k \geq 3$ of $\widehat{\lambda}$, removing c_k gives a contributor $\mu_{k,r+1}$ with $\widehat{\mu}_1 = \widehat{\lambda}_1$ and $\widehat{\mu}_2 = \widehat{\lambda}_2$ unchanged but $\widehat{\mu}_k = \widehat{\lambda}_k - 1$. The threshold becomes

$$\tau(\widehat{\mu}_{k,r+1}) = (q - 1) + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = \tau(\widehat{\lambda}) - 1,$$

giving a prefix bound on T' that is one too weak. The contributor from a row- $k \geq 3$ corner does not match the desired bound on B .

This suggests that either:

1. The threshold formula needs to be refined for $\widehat{\lambda}$ with rows beyond row 2 that have length ≥ 2 ;
2. Or additional structural information (beyond the prefix bound) compensates for the loss in the row- $k \geq 3$ contributor.

This is the natural next-session question.

9.3 Push status

Adding to the eleventh-pass count of 73 unpushed commits: **74 unpushed commits** on `clio-vega/proofs` (after this twelfth-pass commit). The PAT issue from May 6 persists. Robin: please refresh.

10 Status

Established.

- Theorem 8: $B \subseteq E_j^-$ for $1 \leq j \leq \tau(\widehat{\lambda})$ at all two-row $\widehat{\lambda} = (a, b)$, $a \geq b \geq 2$, $q \geq a + b - 2$ — by uniform structural induction.
- As corollaries: the $(5, 4)$, $(6, 3)$, $(6, 2)$, $(6, 4)$, $(5, 5)$, $\dots$ rows of the ladder.

Open.

- Structural sharpness at $j = \tau(\widehat{\lambda}) + 1$ for general two-row $\widehat{\lambda}$.
- Computational verification at new cases (scripts pending).
- Multi-row $\widehat{\lambda}$ (length-preserving corner in row ≥ 3): threshold formula needs refinement or new contributor analysis.
- The dual Frobenius programme (suggested in the May-14 induced-image refutation note).